Scan Detection Report

Network scan analysis from packet captures and Suricata alerts

Report date	8 October 2026 (version 1.0)
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Subject	Five Nmap scans from 192.168.40.129 against 192.168.40.1
Evidence	Five pcapng captures (8 October 2026); Suricata 8.0.7 eve.json and fast.log output; Nmap 7.99 console output
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1. Summary

Five packet captures were analysed together with the Suricata logs generated from them. All five show the same scanner probing the same target across the same 1,000 ports (the Nmap default list), each in two passes. The captures identify each scan type clearly. Suricata did not: none of its alerts is a scan signature.

- **Open service:** only TCP 3580 answered. Nmap identified it as National Instruments LabVIEW service locator httpd 1.0.0. The other 999 ports sent no reply and are filtered, most likely by a host firewall.
- **NULL and XMAS scans and OS detection** revealed nothing usable (Table 2).
- **Suricata alerts:** 9, 68, 13, 10 and 4 for captures 01 to 05 (104 in total). 96 are checksum alerts, most likely caused by NIC checksum offload on the scanner VM. The only alert linked to scan behaviour is SID 2200025, "ICMPv4 unknown code", in capture 03.
- **MITRE ATT&CK:** T1046 and T1595 apply to all five scans; T1592.002 applies additionally to captures 02 and 03 (see Section 4).

Method: the pcapng files were parsed with Python (dpkt) because neither tshark nor Suricata could be installed in the analysis environment. Each Wireshark filter in Section 3 was re-implemented as a separate check and its hit count recomputed from the pcaps; the filters themselves were not executed in Wireshark. The scan-detection rules in Appendix B were run by the report's author with Suricata 8.0.7 on all five captures. The Nmap commands and results are from the attacker's console, a re-run of the same commands after the captures were taken, so scan times differ slightly.

Findings

Ref	Finding	Rating	Basis
F-1	With the default ruleset, none of the five scans triggered a scan-specific Suricata alert. With the local scan rules in Appendix B, all five were detected.	High (before), resolved by rules	104 alerts across the five default runs, none a scan signature (Section 2); Table 6 for the rule results.
F-2	Port 3580 is reachable from the scanner and discloses its product and version in an HTTP banner.	Low to Medium	Nmap -sV: National Instruments LabVIEW service locator httpd 1.0.0; Server header seen in capture 02.

Table 1. Findings. Ratings are the author's assessment for a lab environment and are not a formal scoring method.

2. Scan analysis

Nmap results at a glance

Capture	Command	Nmap result	Nmap time	Probe span in capture
01 SYN	`nmap -sS`	3580/tcp open (nati-svrloc); 999 ports filtered (no response)	13.87 s	15.0 s
02 Service	`nmap -sV`	3580/tcp open, identified as http: National Instruments LabVIEW service locator httpd 1.0.0	25.87 s	13.9 s
03 OS	`nmap -O`	No exact OS match. Nmap found no closed port and gave guesses only: Windows 11 24H2 (92%), Windows 11 21H2 (91%), FreeBSD 6.2 (88%), Windows 10 (86%)	11.02 s	8.8 s
04 NULL	`nmap -sN`	All 1,000 ports open|filtered (no response)	21.30 s	21.1 s
05 XMAS	`nmap -sX`	All 1,000 ports open|filtered (no response)	21.27 s	24.7 s

Table 2. Nmap 7.99 console results, with the probe span measured in the captures. Target: 192.168.40.1 (VMware MAC 00:50:56:C0:00:08, latency 0.2 to 0.7 ms).

2.1 SYN scan (01_syn_scan.pcapng) - nmap -sS

Traffic	1,999 SYN packets (flags 0x002) in 15 s, about 133 packets/s; ports in random order, two passes.
Fingerprint	Window 1024, MSS option only, random TTL and IP-ID per packet, one fixed source port per pass.
Response	One SYN-ACK from port 3580; the scanner replied with RST (half-open). No other replies: 1 open, 999 filtered.
Suricata	All alerts are SID 2200075 (UDPv4 invalid checksum) on background NTP/DNS traffic.

2.2 Service detection (02_service_detection.pcapng) - nmap -sV

Traffic	Same SYN sweep, followed by 10 full TCP connections to port 3580 carrying HTTP requests (GET /, GET /HNAP1, GET /evox/about, POST /sdk, GET /nmaplowercheck...).
Fingerprint	Several requests use the user agent "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)", a direct Nmap indicator. Nmap runs its version-category NSE scripts automatically with -sV, which explains these requests.
Response	11 SYN-ACKs. The target answered with HTTP 503/404 and the header "Server: NI Service Locator/1.0.0 (SLServer)", which is how the service was identified.
Suricata	68 alerts: 44 x SID 2200074 (TCPv4 invalid checksum) on port 3580, 24 x SID 2200075. The TCP alerts coincide with the version probes and are probably offload artefacts; Suricata may have skipped these packets, so HTTP content went uninspected.

2.3 OS detection (03_os_detection.pcapng) - nmap -O

Traffic	SYN sweep plus the fingerprint probe set: TCP probes with unusual flag/window combinations (NULL, SYN+FIN+PSH+URG, FIN+PSH+URG, ECN, SYN with window 31337, ACK), ICMP echo probes (one with code 9), and 300-byte UDP probes to closed ports. About 237 packets/s, the fastest of the five scans.
Fingerprint	Constant IP-ID 4162 on UDP probes; echo requests with code 9 and with IP TOS 0x04.
Response	17 SYN-ACKs only. No reply to the ICMP or UDP probes. Nmap itself warned that it found no closed port, so its OS guesses are unreliable.
Suricata	13 alerts: 8 x SID 2200025 (ICMPv4 unknown code) triggered by the code-9 echo, plus 5 x SID 2200075. The ICMP alerts are the only alerts in the data set caused by the scan itself.

2.4 NULL scan (04_null_scan.pcapng) - nmap -sN

Traffic	2,000 TCP segments with no flags set, window 1024, no options, in 21 s (about 95 packets/s). No replies, so the open port 3580 was not revealed.
Suricata	All alerts are SID 2200075 on background traffic. The 2,000 flows are logged with no flags and state "new".

2.5 XMAS scan (05_xmas_scan.pcapng) - nmap -sX

Traffic	2,000 TCP segments with FIN, PSH and URG set (0x029), window 1024, no options, in 25 s (about 81 packets/s, the slowest). No replies.
Suricata	All alerts are SID 2200075 on background traffic. The engine counted 2,000 URG-flagged packets but no rule used them.

3. Wireshark display filters

Hit counts are packets in captures 01 / 02 / 03 / 04 / 05; a dash means none.

3.1 Scope and context

Filter	Purpose	Hits 01 / 02 / 03 / 04 / 05
`ip.src==192.168.40.129 && ip.dst==192.168.40.1 && tcp`	All scanner-to-target TCP traffic (the scan)	2000 / 2046 / 2080 / 2000 / 2000
`ip.src==192.168.40.1 && ip.dst==192.168.40.129`	Everything the target sent to the scanner; shows how little it answered	5 / 37 / 17 / 4 / 0
`dns.qry.name=="1.40.168.192.in-addr.arpa"`	Reverse DNS lookup of the target, sent just before the first probe of every scan (query and reply)	2 / 2 / 2 / 2 / 2
`ip.src==192.168.40.129 && tcp.window_size_value==1024`	Default Nmap probe window	1999 / 2000 / 2009 / 2000 / 2000

Table 3. Context filters.

3.2 Isolating each scan type

Scan	Filter	What it shows	Hits
SYN	`tcp.flags==0x002`	SYN-only probes	1999 / 2009 / 2021 / - / -
SYN	`tcp.flags==0x012 && tcp.srcport==3580`	SYN-ACK from the open port	1 / 11 / 17 / - / -
SYN	`tcp.flags==0x004 && ip.src==192.168.40.129`	Scanner RST after SYN-ACK (half-open)	1 / 2 / 17 / - / -
Service	`tcp.port==3580 && tcp.len>0`	Version-probe payloads and replies	- / 17 / - / - / -
Service	`http.user_agent contains "Nmap Scripting Engine"`	Nmap NSE signature (request packets)	- / 4 / - / - / -
Service	`frame contains "NI Service Locator"`	Banner of the open service (reply packets)	- / 8 / - / - / -
OS	`icmp.type==8 && icmp.code==9`	Nmap ICMP probe that triggers SID 2200025	- / - / 8 / - / -
OS	`udp.dstport in {34217 41297} && ip.id==0x1042`	Nmap UDP probe (constant IP-ID)	- / - / 8 / - / -
OS	`tcp.flags==0x02b || tcp.flags==0x8c2 || tcp.window_size_value==31337`	SYN+FIN+PSH+URG, ECN and window-31337 probes	- / - / 18 / - / -
NULL	`tcp.flags==0x000`	No flags set	- / - / 8 / 2000 / -
XMAS	`tcp.flags==0x029`	FIN+PSH+URG	- / - / 8 / - / 2000
Any	`tcp.flags.urg==1`	Any URG-bearing probe (rare in normal traffic)	- / - / 16 / - / 2000

Table 4. Filters that isolate each scan type.

4. Comparison of captures and MITRE ATT&CK mapping

"Important alert" is the alert most relevant to the scan. Where Suricata raised nothing scan-related, the row says so and the address columns show the scan traffic itself (marked pcap) alongside the alert traffic.

Capture	Alerts	Important alert	Source IP	Destination IP	Protocol	Port	MITRE ATT&CK
01 SYN scan `nmap -sS`	9: SID 2200075 x9	None scan-related; checksum noise only	192.168.40.129	192.168.40.1 (pcap); alerts: 196.49.100.57, 192.168.40.2	TCP SYN (pcap); alerts: UDP	Top 1,000 ports; open: 3580. Alerts: 123, 53	T1046 Network Service Discovery T1595 Active Scanning
02 Service detection `nmap -sV`	68: SID 2200074 x44, SID 2200075 x24	SID 2200074 TCPv4 invalid checksum (x44)	192.168.40.129	192.168.40.1	TCP (HTTP)	3580	T1046 Network Service Discovery T1595 Active Scanning T1592.002 Gather Victim Host Information: Software
03 OS detection `nmap -O`	13: SID 2200025 x8, SID 2200075 x5	SID 2200025 ICMPv4 unknown code (x8)	192.168.40.129	192.168.40.1	ICMP (alert); TCP and UDP probes (pcap)	ICMP has no port (fast.log "8 -> 9" is type/code). TCP 3580, 43170, 44555; UDP 34217, 41297	T1046 Network Service Discovery T1595 Active Scanning T1592.002 Gather Victim Host Information: Software
04 NULL scan `nmap -sN`	10: SID 2200075 x10	None scan-related; 2,000 flag-less flows not alerted	192.168.40.129	192.168.40.1 (pcap); alerts: 103.97.91.223, 192.168.40.2	TCP, no flags (pcap); alerts: UDP	Top 1,000 ports. Alerts: 123, 53	T1046 Network Service Discovery T1595 Active Scanning
05 XMAS scan `nmap -sX`	4: SID 2200075 x4	None scan-related; 2,000 FIN+PSH+URG probes not alerted	192.168.40.129	192.168.40.1 (pcap); alerts: 103.97.91.223, 192.168.40.2	TCP FIN+PSH+URG (pcap); alerts: UDP	Top 1,000 ports. Alerts: 123, 53	T1046 Network Service Discovery T1595 Active Scanning

Table 5. Comparison of the five captures.

T1595 and T1592.002 fall under Reconnaissance (TA0043); T1046 under Discovery (TA0007). The mappings are the closest official techniques; NULL and XMAS flag abuse has no dedicated technique.

5. Detection gap and recommendations

Suricata loaded 53,098 rules on the first run and 53,103 on the re-run (the extra five are the local rules assessed below). Scanning is a pattern of rate and port diversity, so it needs threshold-based rules rather than content signatures. The ten rules in Appendix B (SIDs 1000201 to 1000210) were written for this purpose and run on all five captures; Table 6 gives the results.

Assessment of the supplied local rules (SIDs 1000101 to 1000105)

The five supplied local rules (SIDs 1000101 to 1000105) loaded without errors and none fired. They look for malware command-and-control behaviour (TLS certificate anomalies, TLS without SNI, IP-check lookups and beaconing; MITRE T1573.002, T1071.001, T1016 and T1029 in the rule metadata). The captures contain no TLS traffic, no IP-check lookup (the only DNS query is a reverse lookup of 192.168.40.1) and no external traffic on port 443, so these rules cannot detect an Nmap scan of an internal host. The scan rules in Appendix B were added for that.

Verified results of the scan-detection rules

SID	Detects	01 SYN	02 Service	03 OS	04 NULL	05 XMAS
1000201	NULL scan, no flags	-	-	-	3	-
1000202	XMAS scan, FIN+PSH+URG	-	-	-	-	3
1000203	SYN sweep, 100+ SYN in 10 s	2	1	1	-	-
1000204	Nmap SYN probe, window 1024	2	1	1	-	-
1000205	ICMP echo code 9	-	-	8	-	-
1000206	SYN+FIN illegal flags	-	-	8	-	-
1000207	UDP probe, IP-ID 4162, 300 bytes	-	-	8	-	-
1000208	Nmap user agent (HTTP)	-	4	-	-	-
1000209	Nmap user agent (raw TCP)	-	4	-	-	-
1000210	Nmap /nmaplowercheck path	-	1	-	-	-
Total		**4**	**11**	**26**	**3**	**3**

Table 6. Alerts per rule and capture, counted from eve.json after running the rules with Suricata 8.0.7 (rules loaded: 10, failed: 0, in all five runs). Console totals: 4, 11, 26, 3 and 3.

- **Every scan is detected.** Each capture raised exactly the rules written for its scan type, and the totals equal the totals predicted beforehand from the packet data.
- **No cross-firing.** The NULL and XMAS rules stayed silent on capture 03, whose 8 NULL-flag and 8 XMAS-flag OS probes are below the threshold of 20. No rule fired on a capture of another scan type.
- **Application-layer rules need checksum checking off.** The user-agent and path rules (1000208 to 1000210) fired in capture 02; in the default runs Suricata recognised no HTTP traffic there, which is consistent with its 44 bad-checksum TCP packets being ignored.

The captures contain almost only scan traffic (one scanner, one target, lab network), so these results show the rules catch the scans but cannot measure false positives on normal traffic; that needs a baseline capture of normal traffic. The SYN rules also match the ECN probe (CWR+ECE+SYN), because those two bits are masked.

Recommendations

- **Checksum noise.** The verified results were obtained with checksum checking off (`-k none`). For lab pcaps, keep it off or disable NIC offload on the capture host; otherwise SIDs 2200074 and 2200075 add noise and may hide the packets you want inspected.
- **Flow-based detection.** Many distinct destination ports from one source within seconds is the clearest indicator and appears in all five captures. A check on the eve.json flow records, or a tool with built-in scan detection, would have caught every scan.

Appendix A. Packet evidence

HTTP exchange from capture 02 (port 3580)

```
Request:  GET /nmaplowercheck1791459243 HTTP/1.1
          Connection: close
          User-Agent: Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
Response: HTTP/1.0 404 Not Found
          Server: NI Service Locator/1.0.0 (SLServer)
          Connection: close
          Content-Type: text/html
```

The request header text is abridged to the fields shown.

Appendix B. Scan-detection rules and how they were run

Rules file: scan_detection.rules (SIDs 1000201 to 1000210). Rate-based rules use a threshold per source address; all other rules match a single packet or request.

Port scans by TCP flag pattern (packet level, threshold per source)

```
alert tcp any any -> $HOME_NET any (msg:"LOCAL SCAN TCP NULL scan (no flags set)"; flags:0; threshold:type both, track by_src, count 20, seconds 10; classtype:attempted-recon; sid:1000201; rev:1; metadata:mitre_tactic_id TA0007, mitre_technique_id T1046;)

alert tcp any any -> $HOME_NET any (msg:"LOCAL SCAN TCP XMAS scan (FIN PSH URG)"; flags:FPU,12; threshold:type both, track by_src, count 20, seconds 10; classtype:attempted-recon; sid:1000202; rev:1; metadata:mitre_tactic_id TA0007, mitre_technique_id T1046;)

alert tcp any any -> $HOME_NET any (msg:"LOCAL SCAN TCP SYN sweep (100+ SYN in 10s from one source)"; flags:S,12; threshold:type both, track by_src, count 100, seconds 10; classtype:attempted-recon; sid:1000203; rev:1; metadata:mitre_tactic_id TA0007, mitre_technique_id T1046;)

alert tcp any any -> $HOME_NET any (msg:"LOCAL SCAN Nmap SYN probe fingerprint (window 1024)"; flags:S,12; window:1024; threshold:type both, track by_src, count 50, seconds 10; classtype:attempted-recon; sid:1000204; rev:1; metadata:mitre_tactic_id TA0043, mitre_technique_id T1595;)
```

Nmap OS detection probes (single packet, no threshold)

```
alert icmp any any -> $HOME_NET any (msg:"LOCAL SCAN Nmap OS detection ICMP echo with code 9"; itype:8; icode:9; classtype:attempted-recon; sid:1000205; rev:1; metadata:mitre_tactic_id TA0043, mitre_technique_id T1592.002;)

alert tcp any any -> $HOME_NET any (msg:"LOCAL SCAN TCP SYN+FIN illegal flag combination (Nmap OS probe)"; flags:SF+,12; classtype:attempted-recon; sid:1000206; rev:1; metadata:mitre_tactic_id TA0043, mitre_technique_id T1592.002;)

alert udp any any -> $HOME_NET any (msg:"LOCAL SCAN Nmap OS detection UDP probe (IP-ID 4162, 300 bytes)"; id:4162; dsize:300; classtype:attempted-recon; sid:1000207; rev:1; metadata:mitre_tactic_id TA0043, mitre_technique_id T1592.002;)
```

Nmap service/version detection and scripts

```
alert http any any -> any any (msg:"LOCAL SCAN Nmap Scripting Engine user agent"; flow:established,to_server; http.user_agent; content:"Nmap Scripting Engine"; classtype:attempted-recon; sid:1000208; rev:1; metadata:mitre_tactic_id TA0043, mitre_technique_id T1595;)

alert tcp any any -> any any (msg:"LOCAL SCAN Nmap Scripting Engine user agent (raw TCP fallback)"; flow:established,to_server; content:"Nmap Scripting Engine"; classtype:attempted-recon; sid:1000209; rev:1; metadata:mitre_tactic_id TA0043, mitre_technique_id T1595;)

alert http any any -> any any (msg:"LOCAL SCAN Nmap NSE probe path /nmaplowercheck"; flow:established,to_server; http.uri; content:"/nmaplowercheck"; startswith; classtype:attempted-recon; sid:1000210; rev:1; metadata:mitre_tactic_id TA0043, mitre_technique_id T1595;)
```

Command used for each capture

```
suricata -r 01_syn_scan.pcapng -c /etc/suricata/suricata.yaml -S scan_detection.rules -k none -l out/01
(same command for 02_service_detection, 03_os_detection, 04_null_scan and 05_xmas_scan, with -l out/02 to out/05)

jq -r select(.event_type=="alert") ... out/01/eve.json | sort | uniq -c  (prints signature_id and signature per alert)
```

The option -S loads only the named rules file, so the default rules and decoder-event alerts do not appear in these results; -k none turns off checksum validation. Hyperscan-cache and unix-socket warnings printed during the runs are harmless for pcap files.